

# How children leave home in the model: the trade-off between the current law and a parent-age law

Decision note for item F3 of the structural review

September 17, 2026

## The object

The model does not track children's ages. It tracks the number of children at home,  $m$ , and applies an exit rule each four-year period. The rule decides how long a child occupies space, hence the space demand of parents by age, which is the margin the paper's mechanism runs on. Two facts frame the choice: the author will not add a child-age state, and the ACS comparison of September 16 showed the current rule leaves 40 percent too few children at home at parent ages 30–42 while keeping “dependents” past 58 who in the data are adult children living with parents.

## The two laws

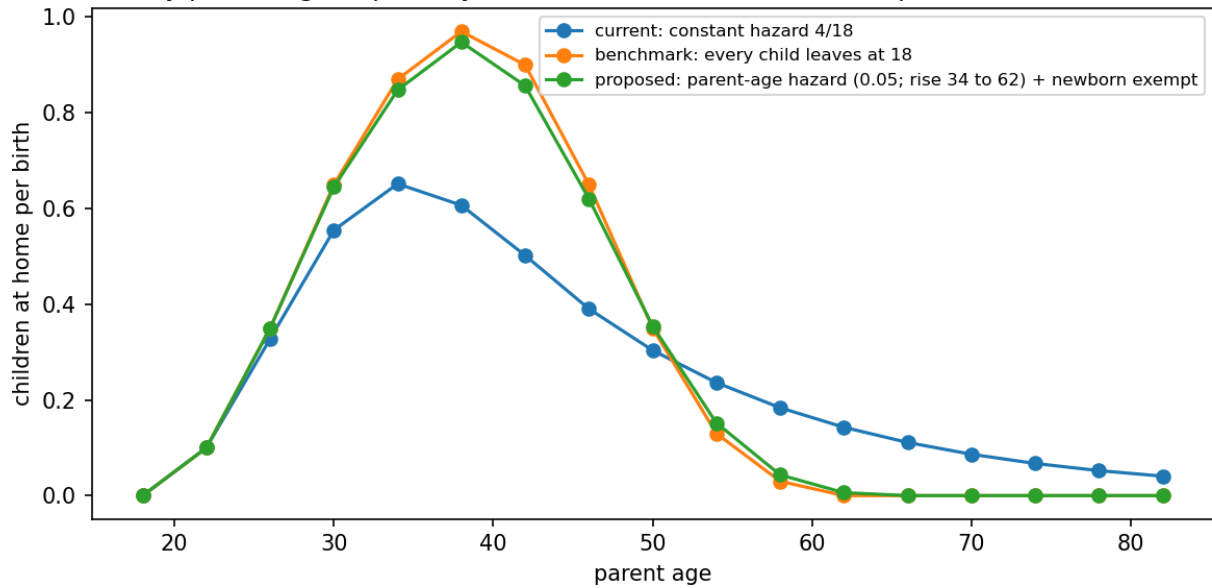
**Current law.** Every child at home leaves with the same probability  $\mu = 4/18 \approx 0.22$  each period, whatever the parent's age and however recently the child was born. Time at home is geometric: no minimum, no maximum. A newborn has a 22 percent chance of being gone by the next date; 8 percent of children are still home after forty years.

**Proposed law.** The exit probability depends on the parent's age  $a$ :  $\mu(a) = \mu_y$  while  $a < a_r$ , rising linearly to one at  $a_f$ , and one after. A child born this period is exempt from this period's draw. Three numbers, chosen once:  $\mu_y = 0.05$ ,  $a_r = 34$ ,  $a_f = 62$ . Still no child state.

## What each law implies

The figure applies each law to one birth per household, with an illustrative distribution of first-birth ages (10 percent at 22, 25 at 26, 30 at 30, 22 at 34, 10 at 38, 3 at 42), and compares with the benchmark in which every child leaves at exactly 18.

ren at home by parent age implied by each maturation law (one birth per household, illustrative



| Law                                         | Years at home per child | Share of child-years while parent is under 46 | Share of child-years at parent age 62+ |
|---------------------------------------------|-------------------------|-----------------------------------------------|----------------------------------------|
| Benchmark: leaves at 18                     | 20.0                    | 0.77                                          | 0.00                                   |
| Current: constant hazard                    | 17.4                    | 0.63                                          | 0.12                                   |
| Proposed: parent-age hazard, newborn exempt | 19.7                    | 0.76                                          | 0.00                                   |

The current law is wrong at both ends at once: it removes children too early (only 63 percent of child-years fall in the parent’s under-46 window, against 77) and too late (12 percent of child-years are at parent ages above 62, against zero). The proposed law matches the benchmark on both counts to within a point. The three parameters were chosen by a grid search on the root mean square distance between the profiles (0.016 for the proposed law against 0.17 for the current one); a rise starting later, at 46, does worse than the current law on the under-46 share, which is why the start is at 34.

## The trade-off, stated honestly

What the proposed law buys: no orphan flow (every child has left before any parent can die, since mortality starts at 66); no adult “dependents”; a young parent’s expected space need in the birth period no longer cut by a fifth; the model’s children-at-home profile can meet the ACS at both ends.

What it costs: the exit clock is the parent’s, not the child’s. A child born to a 22-year-old faces the same low hazard until the parent is 34, then a rising one, so it leaves around the parent’s mid-40s, at 20 to 24 rather than 18; a child born to a 40-year-old leaves between the parent’s 50s and 62, that is, at 12 to 22. The error is largest for the earliest and latest births and zero for births around 30, which is where most first births are. With the illustrative birth ages the average error nets out (19.7 years against 20).

What it does not do: it does not distinguish two children of different ages in the same household; both face the parent's hazard. Under the current law they are also indistinguishable, so nothing is lost relative to now.

## **Interaction with the rest of the model**

Changing the law changes the model counterpart of every moment defined on children at home: the rooms gap between three-plus and one-to-two-child families, the first-birth rooms response, and the childlessness and one-child shares insofar as  $m$  feeds the taste-scale switch. The space floor  $h_P$  and the continuation scale  $\kappa_C$  will move at the refit. Until then, the test is the profile: children at home by parent age, model against ACS, which the sandbox diagnostic already produces.

## **Recommendation**

Adopt the parent-age law with the newborn exemption at (0.05, 34, 62), implemented as a default-off switch (task spec in docs/prompts/TASK\_muse\_maturation\_switch\_20260917.md), and confirm with the ACS profile before any refit. Revisit only if the profile still misses at 30–42, which would say the remaining gap is fertility, not maturation.